

Multiple Choice (10 marks)

1. The Large Magellanic Cloud is ...
 - (a) a dwarf galaxy orbiting the Milky Way.
 - (b) the closest planetary nebula to the Earth.
 - (c) a bright star cluster discovered by Magellan.
 - (d) the outer arm of the Milky Way named after Magellan.
2. Which factor is most important in determining the history of volcanism and tectonism on a planet?
 - (a) size of the planet
 - (b) presence of an atmosphere
 - (c) distance from the sun
 - (d) rotation period
3. Which of the following most likely explains why Venus does not have a strong magnetic field?
 - (a) Its rotation is too slow.
 - (b) It has too thick an atmosphere.
 - (c) It is too large.
 - (d) It does not have a metallic core.
4. The Pleiades is an open star cluster that plays a role in many ancient stories and is well-known for containing ... bright stars.
 - (a) 5
 - (b) 7
 - (c) 9
 - (d) 12
5. Why is the sky blue?
 - (a) Because the molecules that compose the Earth's atmosphere have a blue-ish color.
 - (b) Because the sky reflects the color of the Earth's oceans.
 - (c) Because the atmosphere preferentially scatters short wavelengths.
 - (d) Because the Earth's atmosphere preferentially absorbs all other colors.

True / False (10 marks)

Answer True or False

6. True or False: Helium is the most common element among the Jovian planets, surpassing hydrogen in abundance.
7. True or False: Mars has more extreme seasonal variations because it is farther from the Sun than Earth, resulting in cooler temperatures.
8. True or False: The atmosphere on Mars is too thin to effectively trap a significant amount of heat, limiting the greenhouse effect.
9. True or False: Moons contribute to the stability of particles within rings, gravitational effects at ring edges, and the formation of gaps between rings.
10. True or False: If the angular separation of two stars is small, they will appear as two distinct stars to the naked eye.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the processes and phenomena associated with orbital resonance in celestial mechanics, and critically analyze why the breaking of small Jovian moons does not fall under these causes.
12. Discuss the significance of black bodies in astronomy, providing a detailed definition and exploring how this concept aids in understanding stellar radiation and its implications.

End of Paper